Note to readers: This document shows the approved content, plus my thoughts on graphics and other design elements, labeled "**Design Notes**" to illustrate how I can provide support to the graphic design team.

Rethinking EVs as Battery Storage: Designing Utility Programs for V2G Bidirectional Charging

Introduction and definitions

For decades, America has enjoyed unparalleled grid dependability due in no small part to reliance on "always on" baseload generation that's indifferent to changes in sunlight or wind. In response to regulatory and governmental directives, utilities have turned to renewable energy sources to reduce dependence on fossil fuels and control carbon emissions.

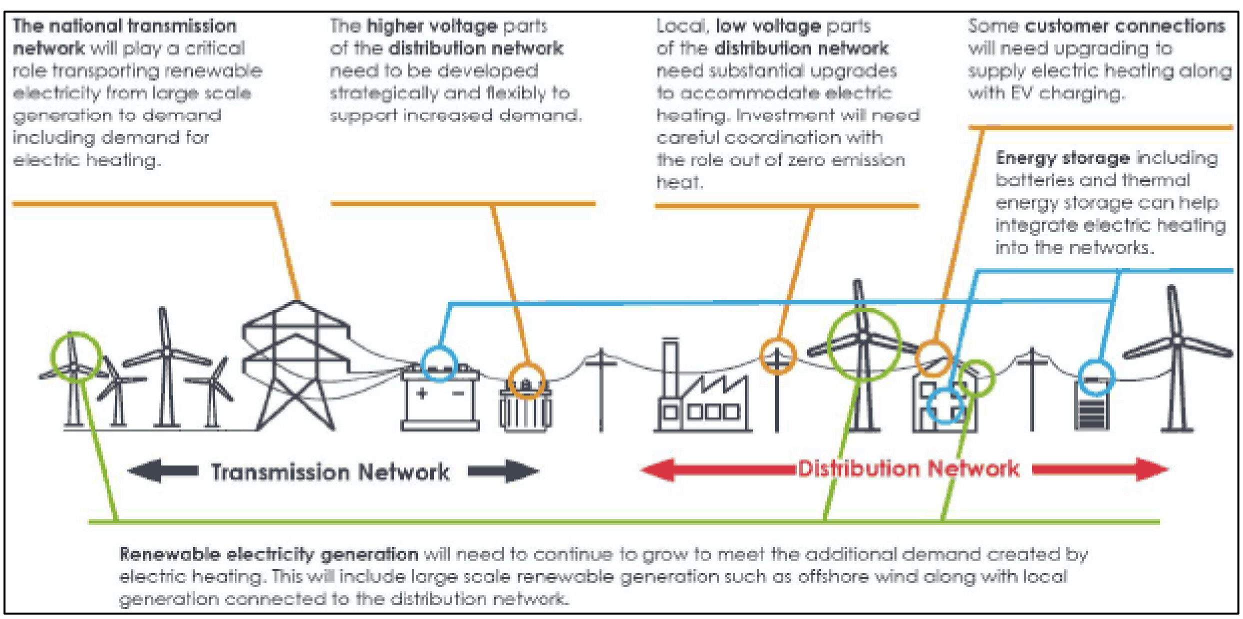
As renewable energy continues to claim a larger share of dispatchable energy portfolios, two new terms are entering the vernacular of utilities and regulators: beneficial electrification and grid resilience.

Beneficial Electrification

The Environmental and Energy Study Institute (EESI) [defines beneficial electrification]() as follows:

> *"**Beneficial electrification** (or strategic electrification) is a term for replacing direct fossil fuel use (e.g., propane, heating oil, gasoline) with electricity in a way that reduces overall emissions and energy costs. There are many opportunities across the residential and commercial sectors. This can include switching to an electric vehicle or an electric heating system – as long as the end-user and the environment both benefit."*

The following graphic illustrates the concept of beneficial electrification.



Design note: Please insert a graphic like this one in this section and highlight the aspect of vehicle batteries (far right side)

Image URL: https://www.gov.scot/binaries/content/gallery/publications/consultation-paper/2021/02/heat-buildings-strategy-achieving-net-zero-emissions-scotlands-buildings-consultation/SCT11205162141_g10.gif

The electric vehicle (EV) is the most visible example of beneficial electrification. Proponents see expanded EV use as a double win: reducing carbon emissions from fossil fuels and eliminating emissions from internal combustion engines in cars and trucks. However, growth in EV use places additional demands on the electric distribution grid, leading to the second new term: grid resilience.

Grid Resilience

One of the Federal government's advanced research laboratories, the Pacific Northwest National Laboratory (PNNL), defines grid resilience this way:

> *"**Grid resilience** in electric utilities is the ability to **avoid or withstand grid stress events** without suffering operational compromise or to **adapt to and compensate** for the resultant strains so as to minimize compromise via graceful degradation. It is, in large part, about what does not happen to the grid or the electricity issue.*
>
> *Grid stress events can include extreme weather and natural disasters, such as wildfires, floods, hurricanes, extreme heat, extreme cold, storms, and any other event that can cause a disruption to the power system."*

PNNL's emphasis is on one-time, finite extreme weather events and natural disasters. However, an influx of EVs with their charging load requirements also challenges grid resilience. How? Utilities engineer their distribution systems to serve a specified load, enabling safe, reliable operation while allowing short periods of overloading up to a specified level. The issue for utilities is that EV loads are not temporary. The distribution system can lose reliability as these loads grow, especially when EVs charge during peak demand.

Design note: Please insert a graphic of a typical electric distribution system in this section.

The Opportunity for EVs as Distributed, Dispatchable Battery Storage

The influx of EV loads can become a grid-resilience asset rather than a detriment when the utility supports the aggregation and dispatch of these grid-edge distributed energy sources through vehicle-to-everything (V2X) bidirectional charging. V2X includes vehicle-to-grid (V2G) and vehicle-to-building (V2B).

The V2X approach benefits both the utility and the EV owner in the following ways:

V2X Participants	Revenue from energy sales	Savings from reduced energy use	Reduced demand during peaks	Increased, rapid response power supply	Faster ROI from EV investment	Reduced investment in the local distribution system load upgrades	
Fleet owner	X	X			X		
Utility			X	X		X	

Design note: Please insert a graphic here that shows the value flow in both energy and dollars for a typical V2X situation.

Utilities should view an aggregation of EVs as providing the same power resource as utility-scale batteries (defined by the US Energy Information Agency as one megawatt (MW) or larger), except that, in the V2X setting, the utility does not need to buy, install, and maintain these large-capacity batteries.

Requirements for a Successful V2X Program Implementation

Implementing a V2X program doesn't have to be difficult, especially when the utility already has programs for peak-period load control, demand response, and interconnection of distributed renewable energy resources, most commonly solar. Let's examine the nuances of building, deploying, and managing a V2X program.

Utility Requirements

There are five fundamental elements needed to implement a V2X program. They are:

- Regulatory approval and support.
- Interconnection standards, including an approval process and standard interconnection engineering design.
- Valuation of the EV energy output based on prevailing market rates, avoided cost of energy, time of day, etc.
- A dynamic hourly rate detailing energy value, performance requirements, and penalties for non-performance.
- Software and hardware to manage communication with the connected EVs, control dispatch, and track energy purchased, including these utility systems:
    - Billing/Accounting
    - SCADA
    - Interconnected EV monitoring and control (dispatch, demand response)
    - Meter reading
    - Power supplier (RTO, ISO, Utility-owned) communications for price signals, energy use
    - Connected load communications

Design note: Please insert a generic interconnection design in this section.

Transactive Rates – The New Hampshire Electric Cooperative (NHEC) Experience

NHEC has created a dynamic pricing structure for V2X use called a transactive energy rate (TER). At its core, it is a dynamic hourly rate that receives price signals (locational marginal price – LMP) from NHEC's power supplier, ISO New England. The ISO forecasts rates a day in advance, and NHEC makes them available the afternoon of the day prior.

Before getting into the control mechanics, let's look at NHEC's TER process. Customers with a Nissan LEAF or a Generac PWRcell electric storage battery, connected to a Fermata Energy charger, register their vehicle or battery with the program. This registration instructs NHEC's metering and accounting systems to configure the car or battery as a virtual meter, tracking the bidirectional flow of energy.

Now, for the mechanics of the TER. The day-ahead hourly forecast is communicated to the Fermata Energy platform using the Open Automated Demand Response (OpenADR) protocol. The Fermata Energy management software combines the forecast with hundreds of other data points to calculate the optimal times to sell energy to the grid, use it in the facility, charge the battery, prevent charging, or take no action. Fleet owners know in real time how to respond to maximize the economic value of their assets. The choice to participate is always up to the fleet operator, subject to the rate.

Design note: If NHEC has a graphic of how their rate works or something else interesting, please use it here. Also, a screenshot of the Fermata Energy software would be appropriate.

EV Fleet Owner Requirements

For the owners of an EV fleet, the requirements are more straightforward. First and most importantly, the fleet owner must confirm that participation doesn't violate the EV manufacturer's battery warranty.

Once that's determined, apart from corporate approvals to proceed, fleet owners will need to purchase, install, and maintain the appropriate number of Fermata Energy chargers and install the management software client on their network. In addition, the fleet owner must develop a process to support responding to the system's real-time signals.

Design note: Include a graphic of the ConEd Revel program or the V2B program in Boulder, CO.

Closing Thoughts

EV penetration rates will continue accelerating, impacting local distribution grid resiliency, first at the grid edge and later upstream. We encourage all electric utilities, whether investor-owned, cooperative, or municipal, to turn EVs into power supply assets. Fermata Energy's hardware and market-leading software enable utilities with the most complex rate structures and requirements to implement a successful V2X program.

Insert standard CTA here